



24 JUNE 2026

ITC · SHARING SESSION · DATA SCIENCE & FINANCIAL ENGINEERING

From Classroom to Career

A Data Science graduate's journey into **Actuarial Science** & **Business Intelligence**

PRESENTED BY

SONG Phalla

Business Intelligence Team Lead · Goodhill Enterprise

Former Actuarial Analyst · AIA Cambodia

From Intern to Team Lead in Under Two Years



SONG Phalla

BI Team Lead · Goodhill

- First-generation graduate of ITC's Data Science program
- ~1.5 years across insurance & retail analytics
- Path: intern → specialist → team lead



JUN–AUG 2024 (3 months)

Data Analyst Intern



Mitsubishi Motor Cambodia



JAN–DEC 2025 (1 year)

Actuarial Analyst



AIA Cambodia · Life insurance



JAN–MAR 2026 (3 months)

Senior Business Intelligence



Goodhill Enterprise



APR 2026 – NOW (Promoted)

Business Intelligence Team Lead



Goodhill Enterprise

What Is an Actuary?

An actuary uses mathematics, statistics, and financial theory to measure and manage risk — in short, to **put a price on an uncertain future.**



Pricing

Set the premium a customer pays for a policy.



Reserving

Set aside money today to pay future claims.



Risk Management

Protect the company from large, rare losses.



Experience Studies

Analyze real data: deaths, claims, lapses.

WHERE ACTUARIES WORK

Life & health insurers (like AIA), general insurance, reinsurance, pension funds, consulting firms, and banks.



Maps directly to ITC's new **Financial Engineering major**

Skills You Need

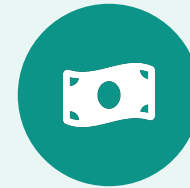
And where you already started learning them at ITC.



Mathematics & Probability

Expected values, distributions, mortality rates.

Probability · Statistics



Financial Mathematics

Time value of money, discounting, annuities.

Financial Math



Programming & Data

Excel, SQL, R/Python

Programming · Databases



Business & Communication

Explain technical results to non-technical people.

Soft skills



Reality check: about 70% of the job is data, assumptions, and careful modeling — the maths from ITC is your foundation, not the whole job.

How Actuaries Price a Product

THE EQUIVALENCE PRINCIPLE

$$\text{EPV(Premiums)} = \text{EPV(Benefits)} + \text{EPV(Expenses)} + \text{Profit}$$

EPV = Expected Present Value — today's value of uncertain future cash flows.



Two ideas drive everything: (1) the probability the claim actually happens, and (2) discounting — money today is worth more than the same money tomorrow.

Pricing in Action: A Simple Example



1-Year Term Life Policy

Customer age

30

Payout if death (Sum Assured)

\$10,000

Chance of death this year

0.1% (0.001)

Interest / discount rate

5%

** Illustrative figures, for teaching.*

1

Step 1 · Expected claim this year

$$0.001 \times \$10,000 = \$10$$

2

Step 2 · Discount to today

$$\$10 \div 1.05 = \$9.52$$

Net premium

3

Step 3 · Add expenses + profit

$$\$9.52 + \$5 + \$2 \approx \$16.52$$

Gross premium

Customer pays **≈ \$16.52 today** for **\$10,000** of cover.

Price = expected cost of risk + time value of money + expenses + profit

The SOA Exam Path

To qualify, actuaries pass a series of professional exams. The main global body is the **SOA — Society of Actuaries (USA)**. Others: IFoA (UK) and CAS (general insurance).

START HERE



7–10

exams in total

4–7+

years (while working)

~300h

study per exam

Sponsored

fees + study leave



Start with Exam P & FM — they overlap heavily with your ITC Probability and Financial Math courses. Many employers (like AIA) sponsor exam fees and give study leave.

My Current Role: BI Team Lead

Business Intelligence (BI) turns raw company data into **clear insights** that leaders use to make decisions.

What I do as BI Team Lead



Build dashboards & reports — sales, inventory, finance



Design data pipelines & a single source of truth



Partner with teams: what's happening, and why?



Lead & mentor the BI team — set standards & priorities

SQL

Power BI

Excel

Python

Problem solving

Storytelling

Actuarial vs BI



Actuarial

Deep, specialized risk & pricing. Insurance only.



Business Intelligence

Broad, cross-functional decision support. Any industry.

Leadership lesson: technical skill gets you in the door; communication & ownership move you up.

Key Takeaways for Juniors

-  Your ITC foundation — maths, stats, coding — is in real demand.
-  Internships matter. My 3-month internship started everything.
-  Two strong paths: Financial Engineering (actuarial) or Data/BI.
-  Eyeing actuarial? Start SOA Exam P & FM early — they overlap your courses.
-  Master tools — Excel, SQL, Power BI, Python travel across any job.
-  Soft skills win — learn to explain complex things simply.

“From first-generation student to team lead in under two years — you can too.”

Thank you! Questions & Answers · phollapholla715@gmail.com

